

Chapter 2, Data Analysis and Visualization

The purpose to this notebook is to create visualization for the 2016 data.

Import python libraries

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib as mpl
import re
from matplotlib import pyplot as plt

%matplotlib inline
```

Import the data

```
In [2]: totals65 = None
totals65 = pd.read_csv('T65_Totals.csv')
```

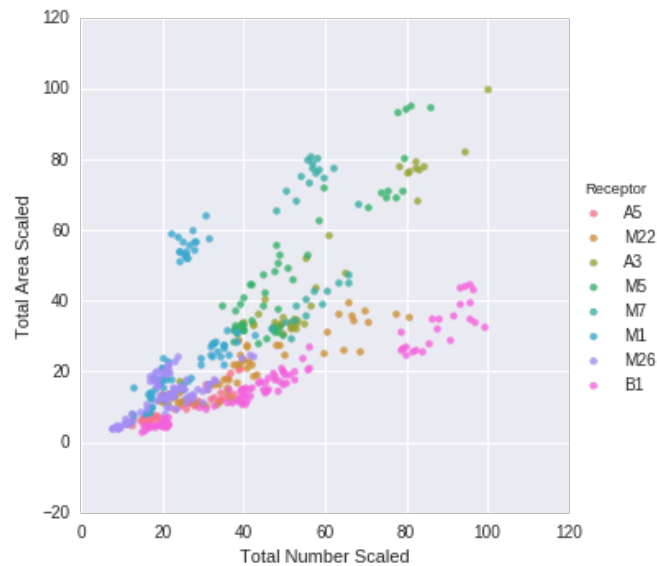
```
In [3]: totals125 = None
totals125 = pd.read_csv('T125_Totals.csv')
```

Visualizations

Unless otherwise noted, the graphs will use the scaled values.

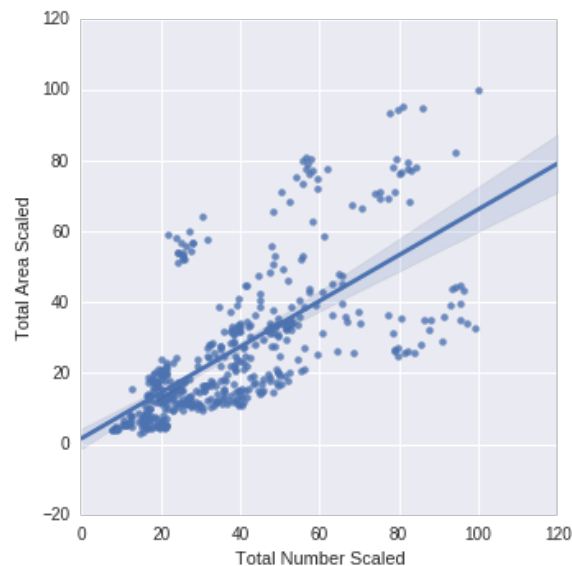
Total Number x Total Area -- All wild and mutants

```
In [4]: number_by_area_all = sns.lmplot('Total Number Scaled', 'Total Area Scaled',
                                         data=totals65,
                                         fit_reg=False,
                                         hue='Receptor')
number_by_area_all.savefig('total65_number_by_area_all.png')
```



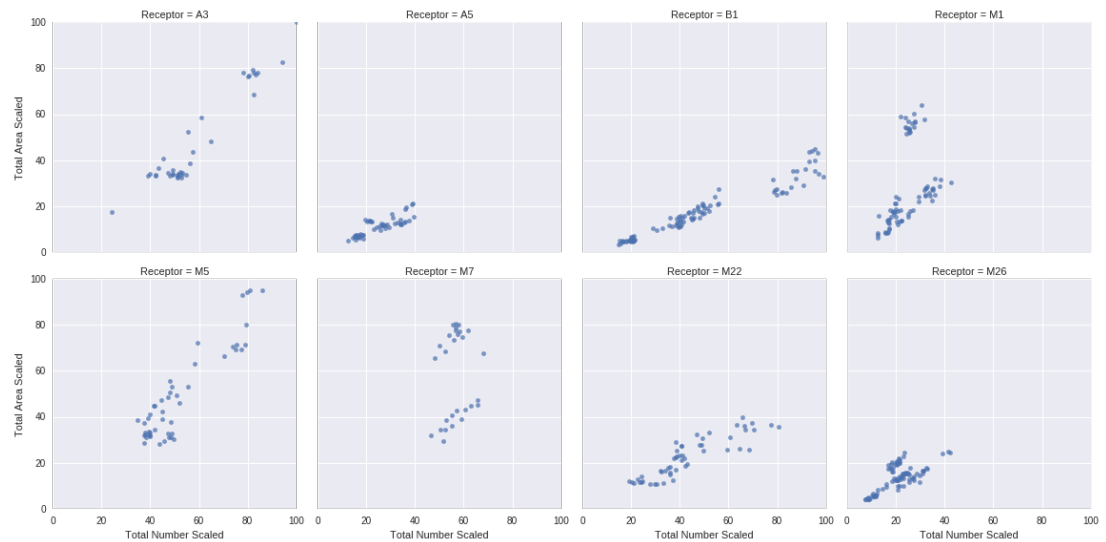
Total Number x Total Area -- All wild and mutants with regression line

```
In [5]: number_by_area_all_regression = sns.lmplot('Total Number Scaled', 'Total Area Scaled',
                                                    data=totals65,
                                                    fit_reg=True)
number_by_area_all_regression.savefig('total65_number_by_area_all_regression.png')
```



Individual Receptors, All steps then by individual step

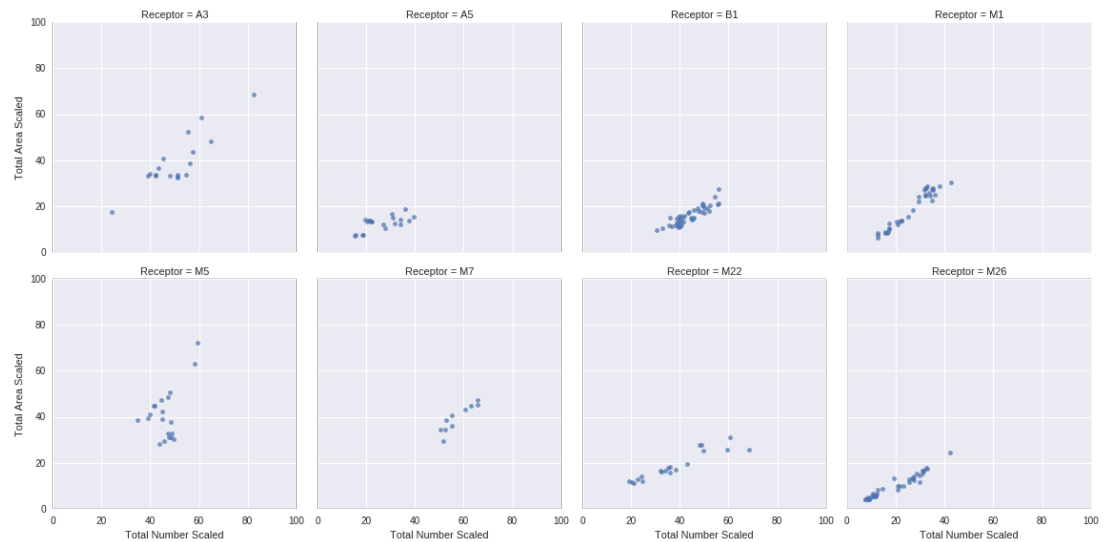
```
In [6]: g = sns.FacetGrid(totals65, col='Receptor', col_wrap=4, size=4, ylim=(0,100)
, xlim=(0,100),
                                col_order=['A3', 'A5', 'B1', 'M1', 'M5', 'M7', 'M22', 'M26'])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.add_legend()
g.savefig('total65_grid_by_receptor.png')
```



STEP: -l-post

```
In [7]: tmpdf = totals65.loc[totals65['Experiment Step'] == '-l-post']

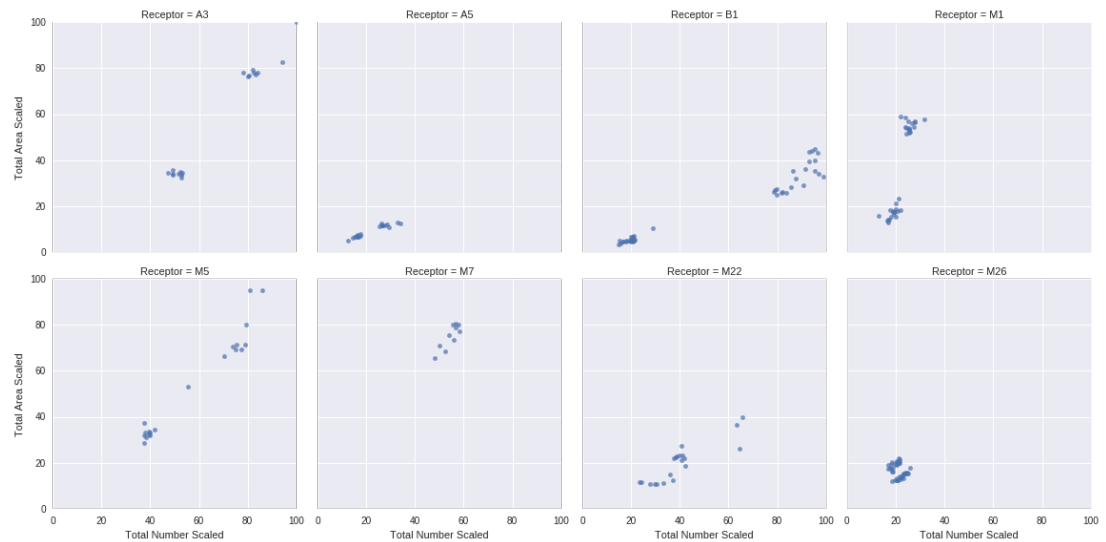
g = sns.FacetGrid(tmpdf, col='Receptor', col_wrap=4, size=4, ylim=(0,100), x
lim=(0,100),
                col_order=['A3', 'A5', 'B1', 'M1', 'M5', 'M7', 'M22',
', 'M26'])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.add_legend()
g.savefig('total65_grid_by_receptor_ligand_post.png')
```



STEP: -nl-post

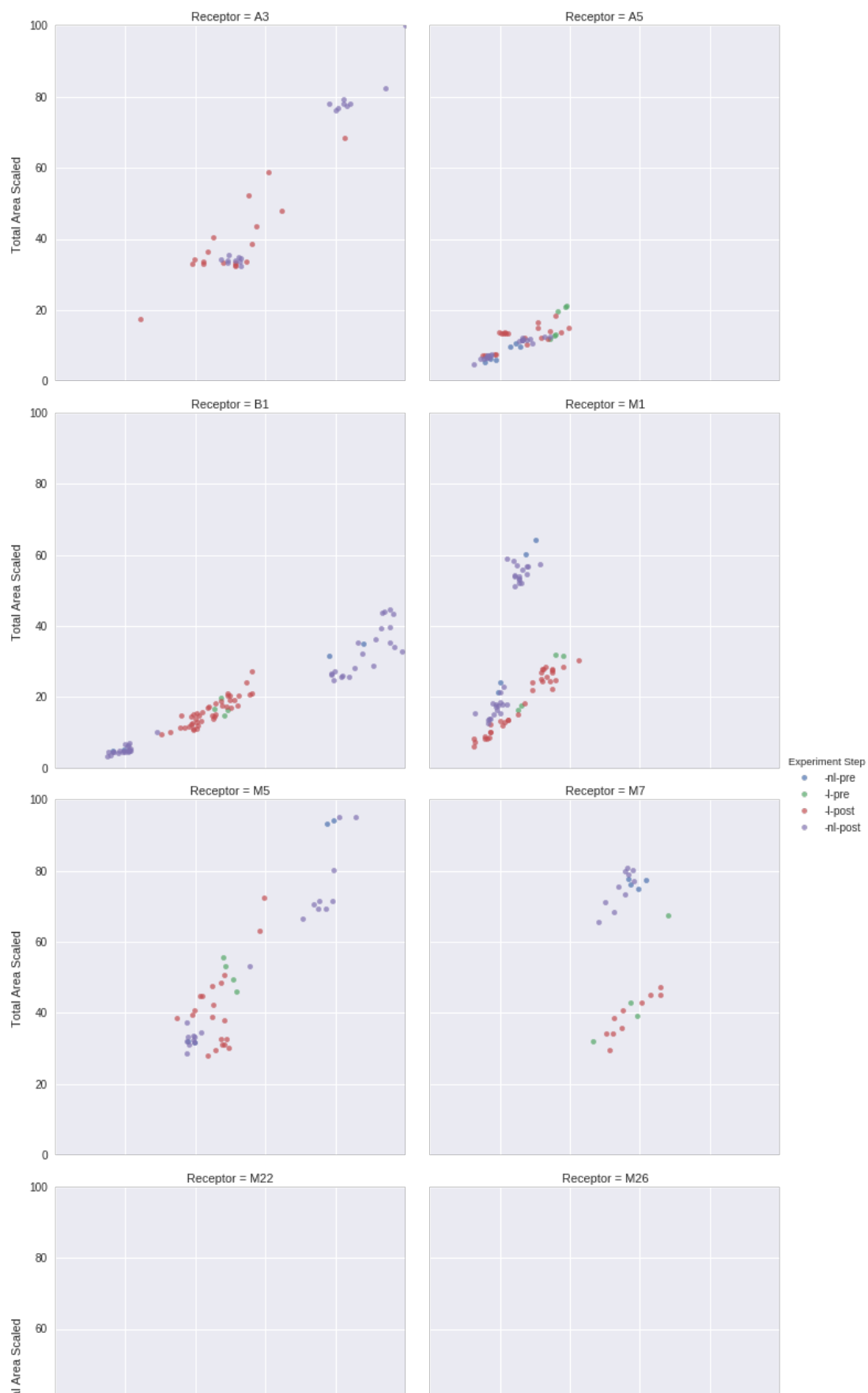
```
In [8]: tmpdf = totals65.loc[totals65['Experiment Step'] == '-nl-post']

g = sns.FacetGrid(tmpdf, col='Receptor', col_wrap=4, size=4, ylim=(0,100), x
lim=(0,100),
                  col_order=['A3', 'A5', 'B1', 'M1', 'M5', 'M7', 'M22',
                              'M26'])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.add_legend()
g.savefig('total65_grid_by_receptor_noligand_post.png')
```



Individual Receptors, all steps in each graph

```
In [9]: g = sns.FacetGrid(totals65,
                          col='Receptor',
                          col_wrap=2,
                          size=5,
                          hue='Experiment Step',
                          ylim=(0,100), xlim=(0,100),
                          aspect=1,
                          col_order=['A3', 'A5', 'B1', 'M1', 'M5', 'M7', 'M22', 'M26
                          '])
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled', alpha=.7)
g.add_legend()
g.savefig('total65_individual_receptors_all_steps.png')
```

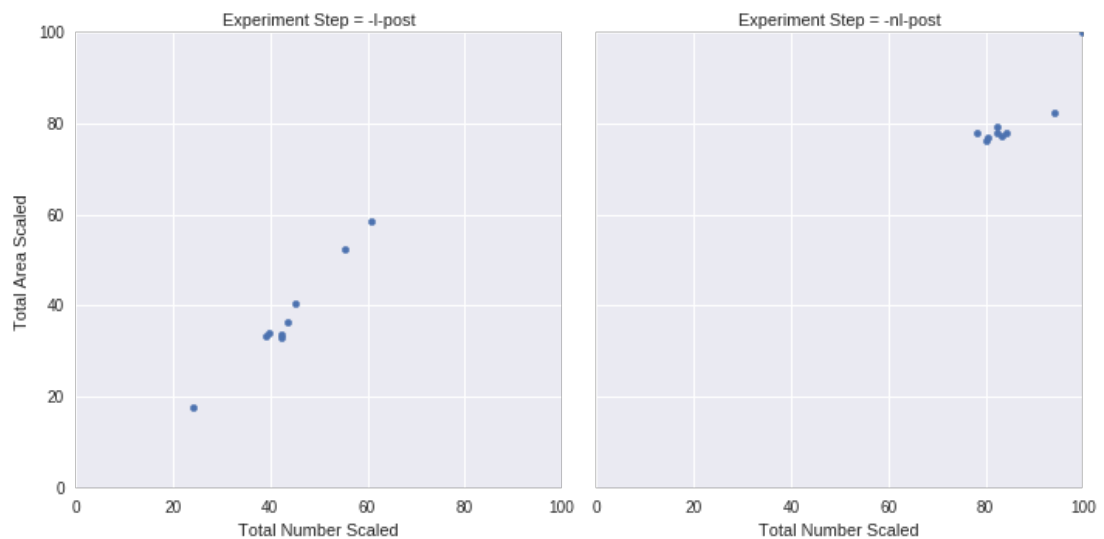


Receptor A3 - post steps, run group 1

```
In [10]: tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'] == 'A3') &
                              (totals65['Run Group'] == 1)])

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.savefig('total65_a3_post-steps_run-group-1.png')
```

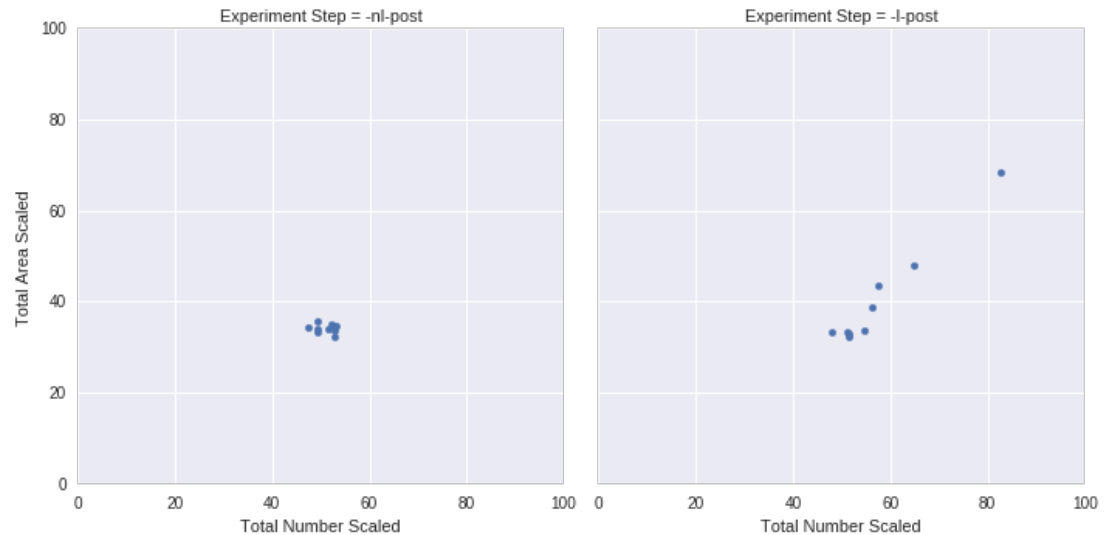


Receptor A3 - post steps, run group 2

```
In [11]: tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'] == 'A3') &
                             (totals65['Run Group'] == 2)])

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.savefig('total65_a3_post-steps_run-group-2.png')
```



Receptor A3 post steps, both run groups

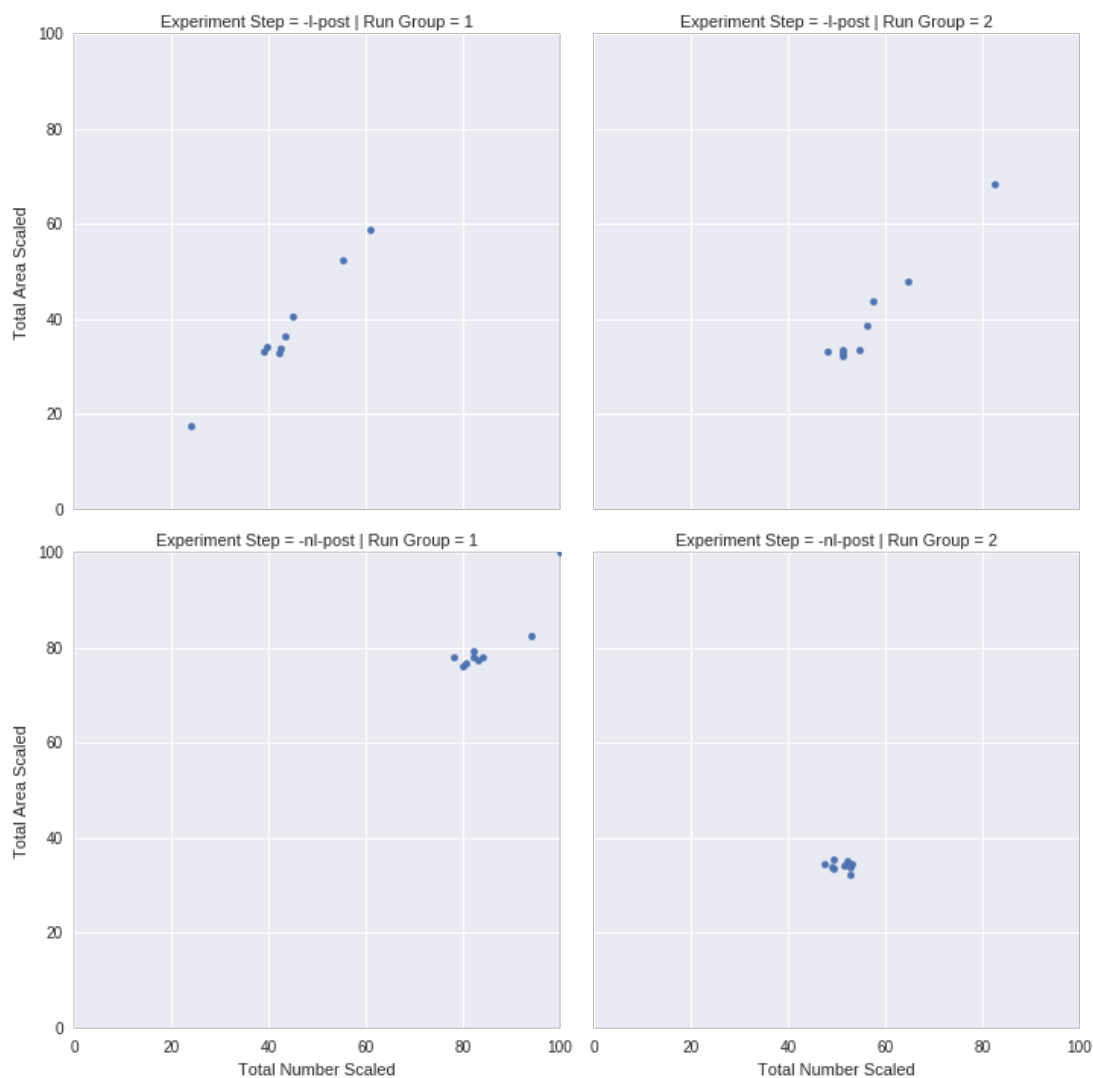
```

In [12]: post = ['-l-post', '-nl-post']
tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'] == 'A3') &
                               (totals65['Experiment Step'].isin(post))
                               ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('total65_a3_post-steps_both-run-groups.png')

```



Receptor A5 - post steps, run group 1

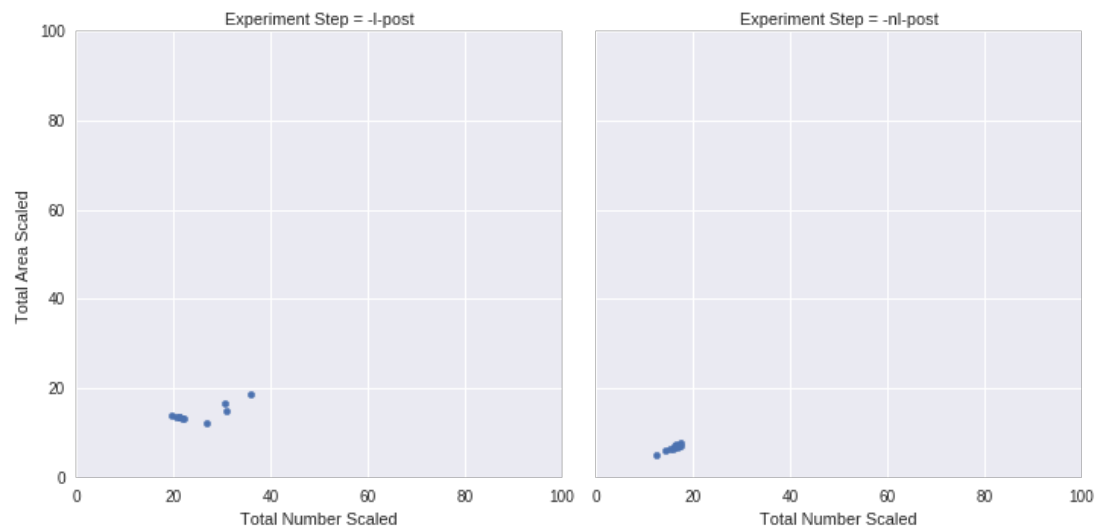
```

In [13]: post = ['-l-post', '-nl-post']
tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'] == 'A5') &
                              (totals65['Run Group'] == 1) &
                              (totals65['Experiment Step'].isin(post))
                              ])

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('total65_a5_post-steps_run-group-1.png')

```



Receptor A5 - post steps, run group 2

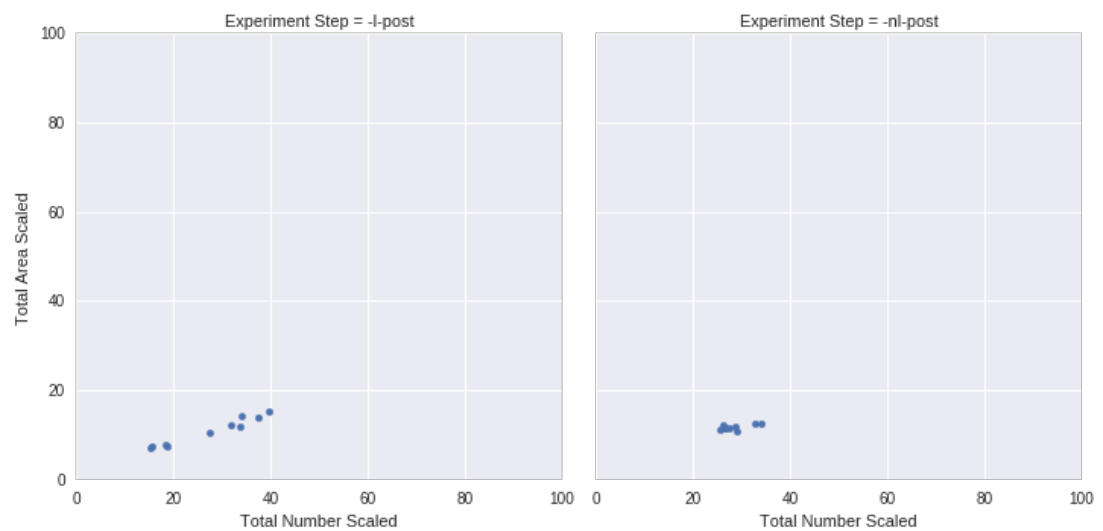
```

In [14]: post = ['-l-post', '-nl-post']
tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'] == 'A5') &
                              (totals65['Run Group'] == 2) &
                              (totals65['Experiment Step'].isin(post))
                              ])

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('total65_a5_post-steps_run-group-2.png')

```



Receptor A5 post steps, both run groups

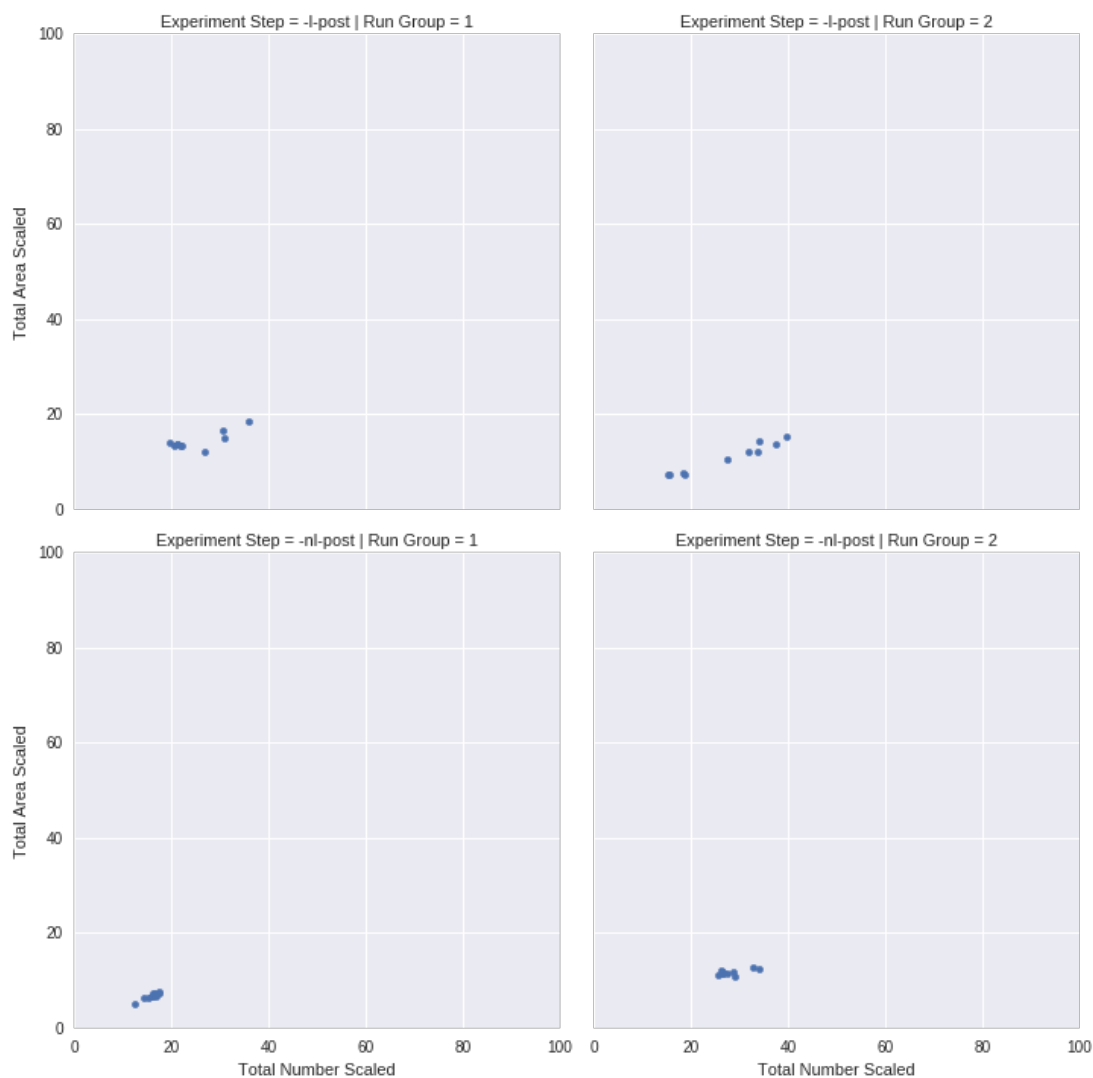
```

In [15]: post = ['-l-post', '-nl-post']
tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'] == 'A5') &
                             (totals65['Experiment Step'].isin(post))
                             ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('total65_a5_post-steps_both-run-groups.png')

```



Receptor A3 and mutants, post steps, both run groups

```

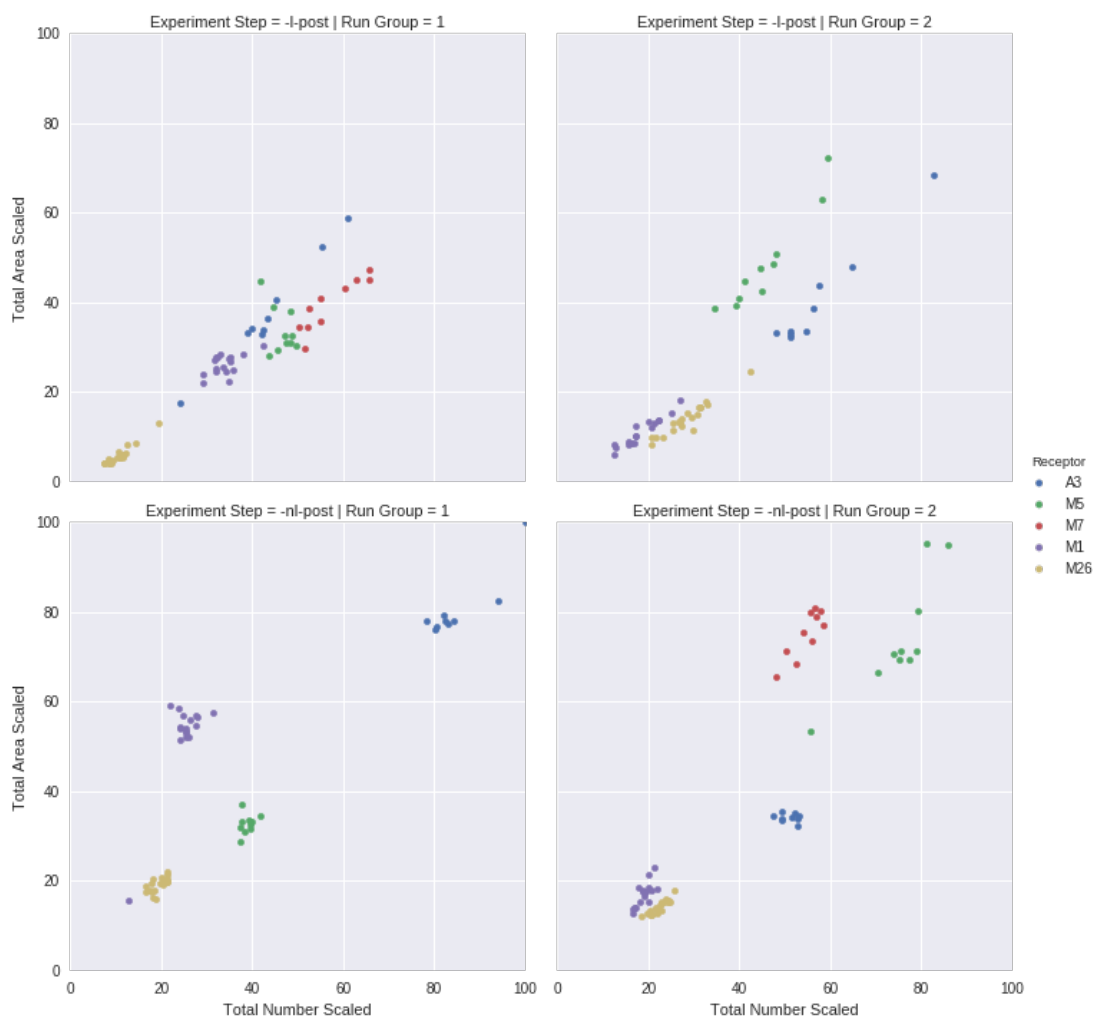
In [16]: post = ['-l-post', '-nl-post']
receptors = ['A3', 'M1', 'M5', 'M7', 'M26']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                              (totals65['Experiment Step'].isin(post))
                              ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('total65_a3-and-mutants_post-steps_both-run-groups.png')

```



Receptor A5 and mutants, post steps, both run groups

```

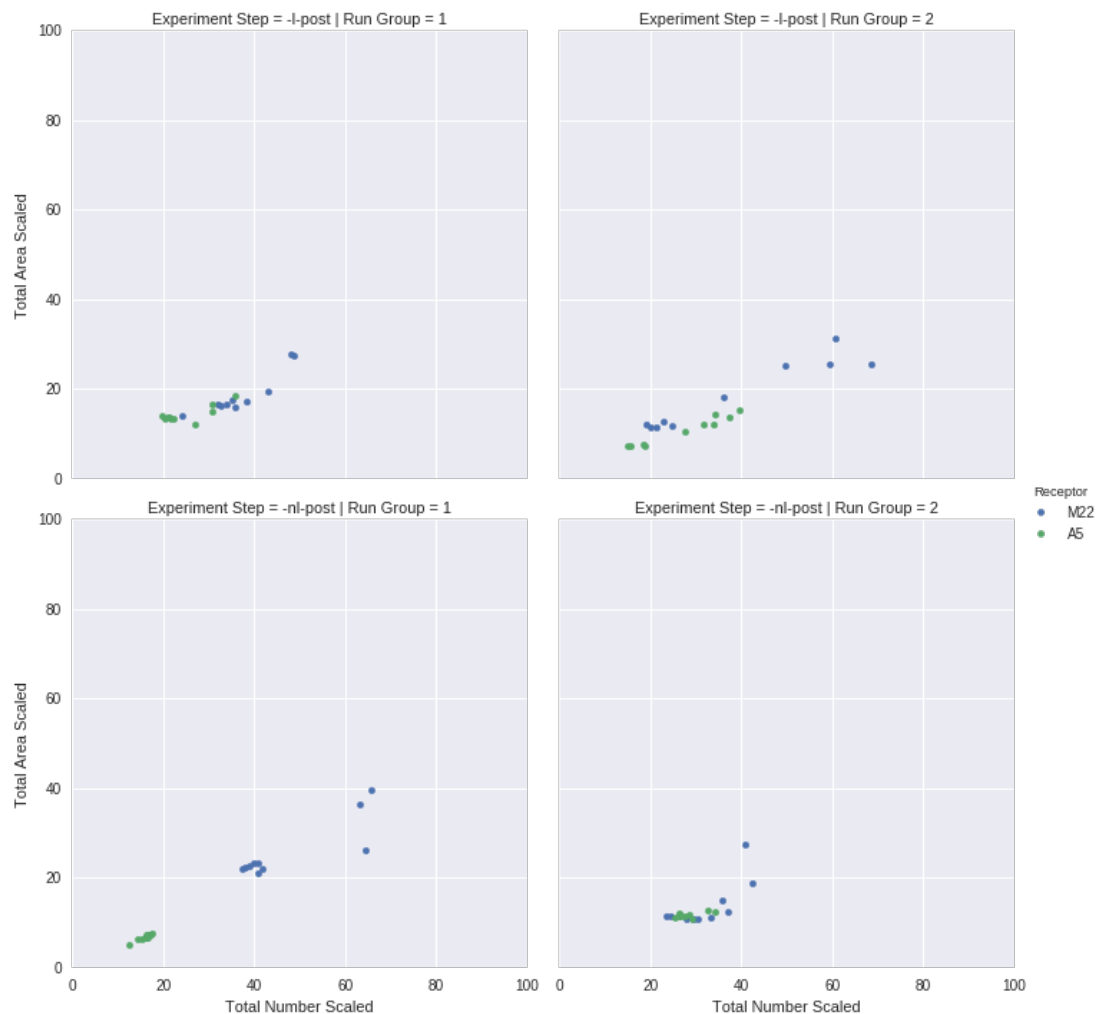
In [17]: post = ['-l-post', '-nl-post']
receptors = ['A5', 'M22']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                               (totals65['Experiment Step'].isin(post))
                               ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('total65_a5-and-mutants_post-steps_both-run-groups.png')

```



Receptor A3 and mutants, post steps, time series for area, both run groups

```

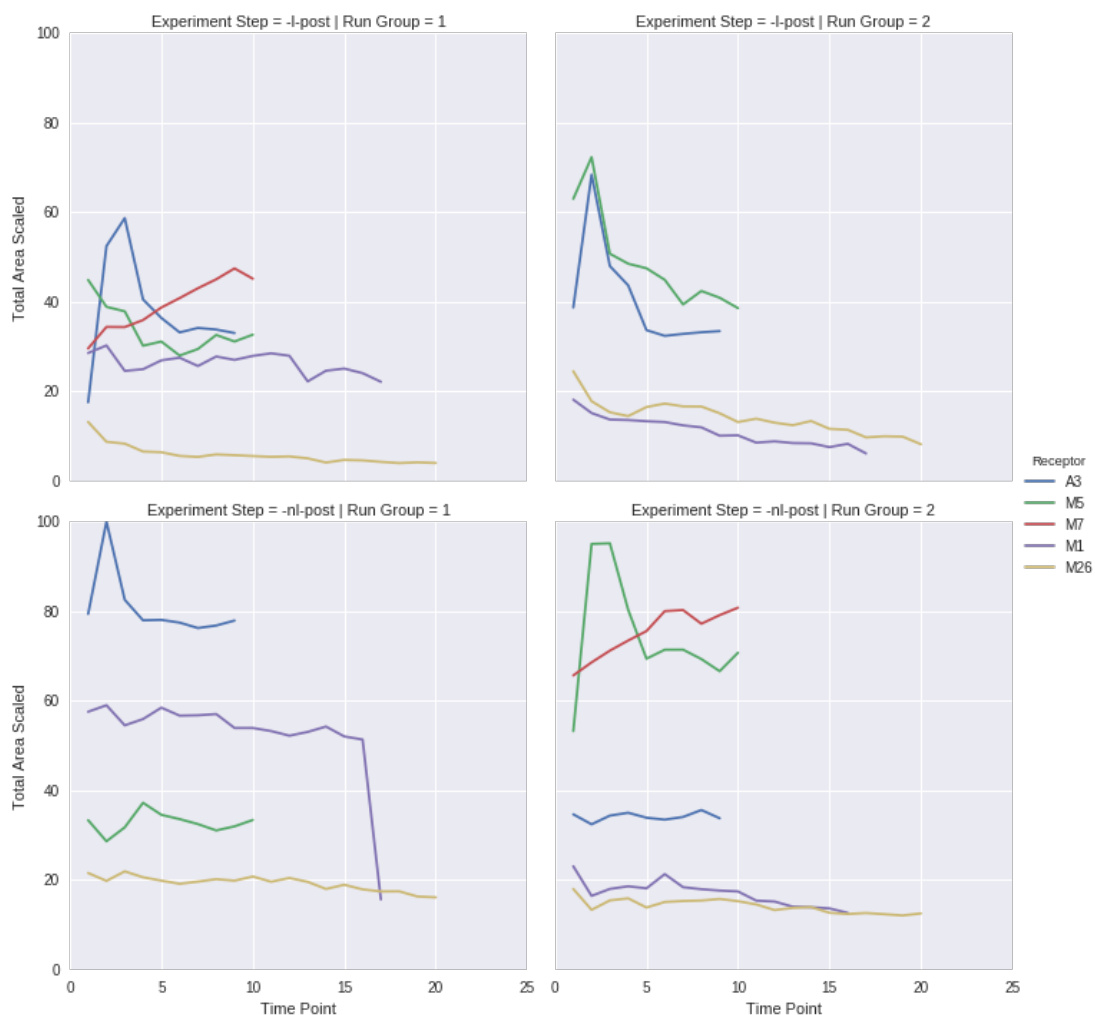
In [18]: post = ['-l-post', '-nl-post']
receptors = ['A3', 'M1', 'M5', 'M7', 'M26']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                               (totals65['Experiment Step'].isin(post))
                               ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,25),
                  aspect=1)

g.map(plt.plot, 'Time Point', 'Total Area Scaled')
g.add_legend()
g.savefig('total65_a3-and-mutants_post-steps_time-series-for-area_both-run-g
roups.png')

```



Receptor A3 and mutants, post steps, time series for number, both run groups

```

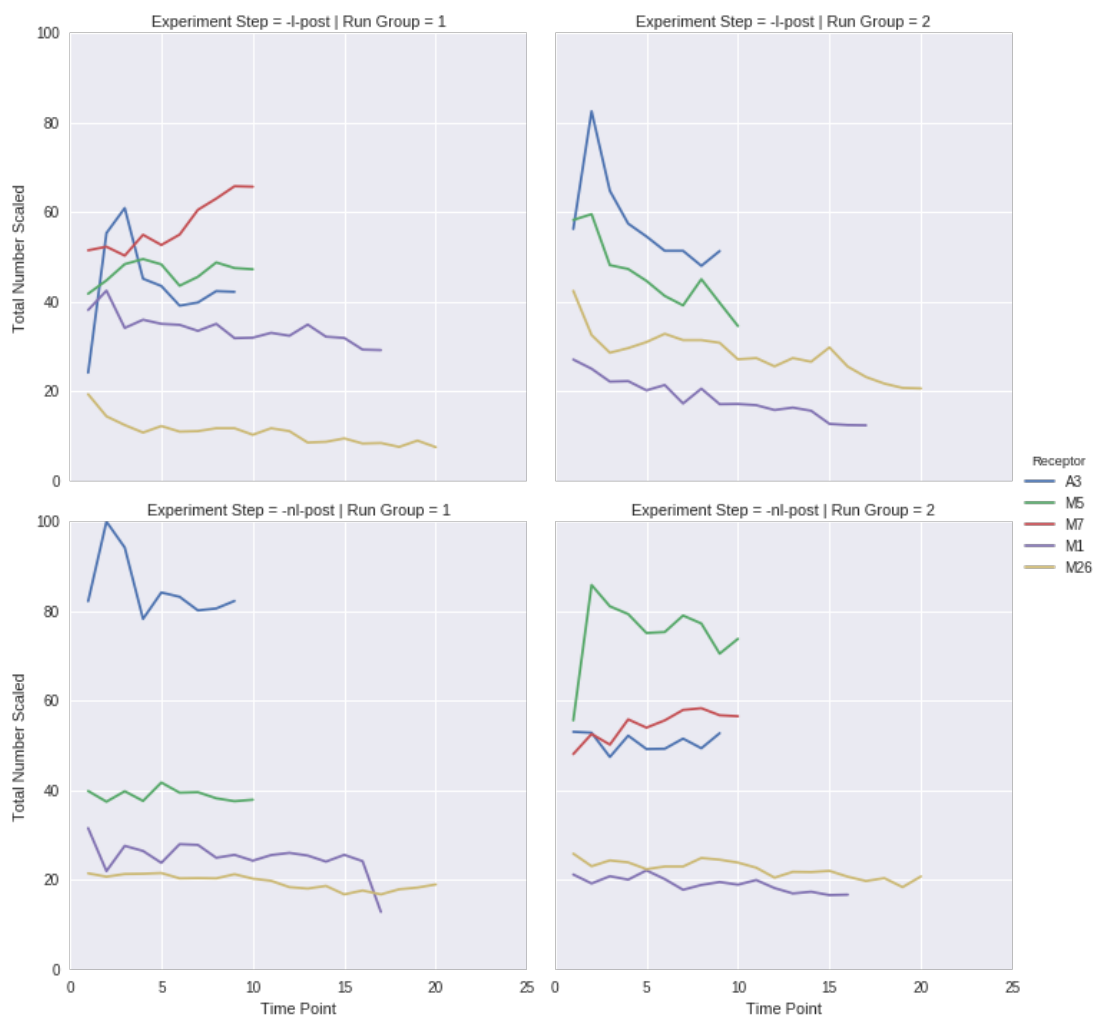
In [19]: post = ['-l-post', '-nl-post']
receptors = ['A3', 'M1', 'M5', 'M7', 'M26']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                               (totals65['Experiment Step'].isin(post))
                               ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,25),
                  aspect=1
                  )

g.map(plt.plot, 'Time Point', 'Total Number Scaled')
g.add_legend()
g.savefig('total65_a3-and-mutants_post-steps_time-series-for-number_both-run-
-groups.png')

```



Receptor A3 and mutants, post steps, time series for intensity, both run groups

```

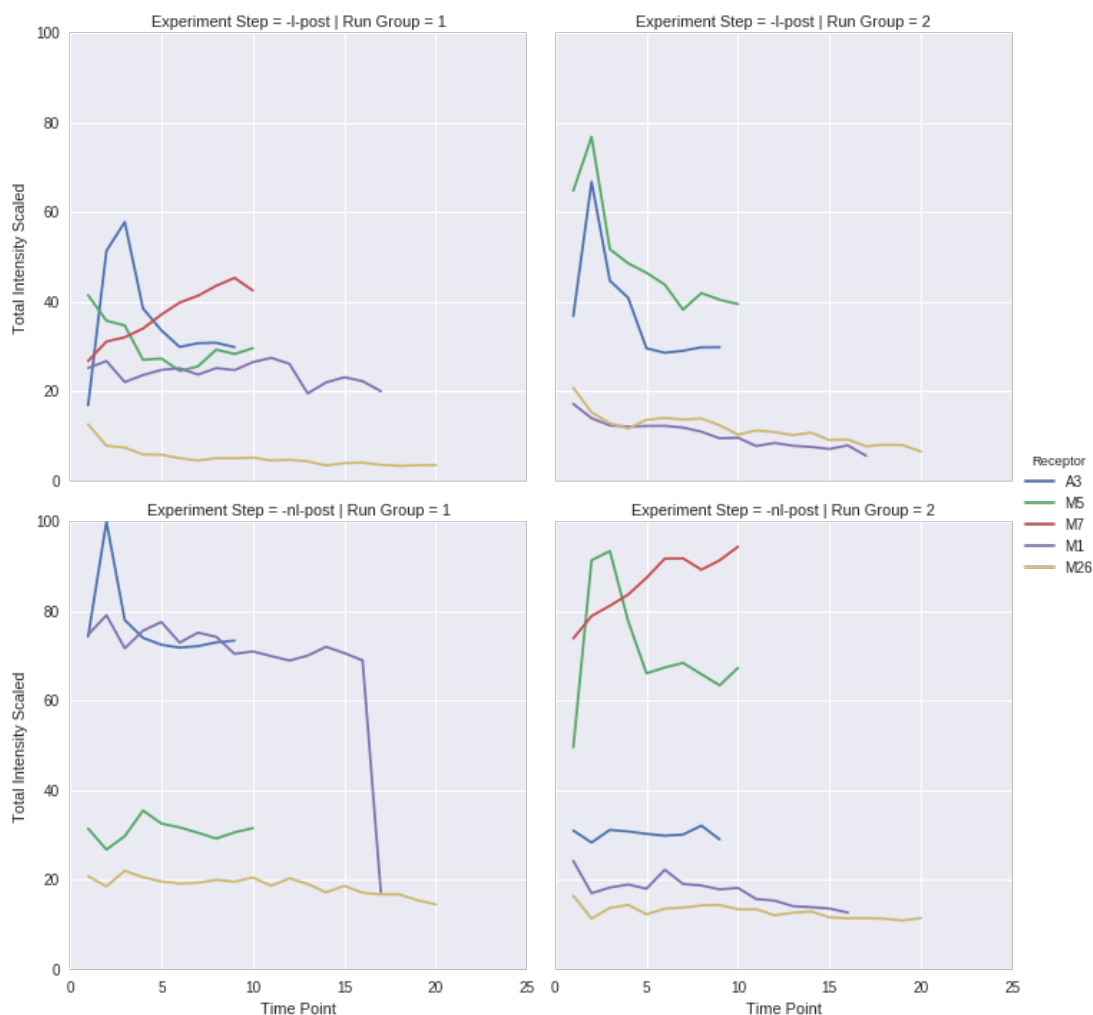
In [20]: post = ['-l-post', '-nl-post']
receptors = ['A3', 'M1', 'M5', 'M7', 'M26']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                               (totals65['Experiment Step'].isin(post))
                              ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,25),
                  aspect=1)

g.map(plt.plot, 'Time Point', 'Total Intensity Scaled')
g.add_legend()
g.savefig('total65_a3-and-mutants_post-steps_time-series-for-intensity_both-
run-groups.png')

```



Receptor A3 and M1, post steps, time series for area, both run groups

```

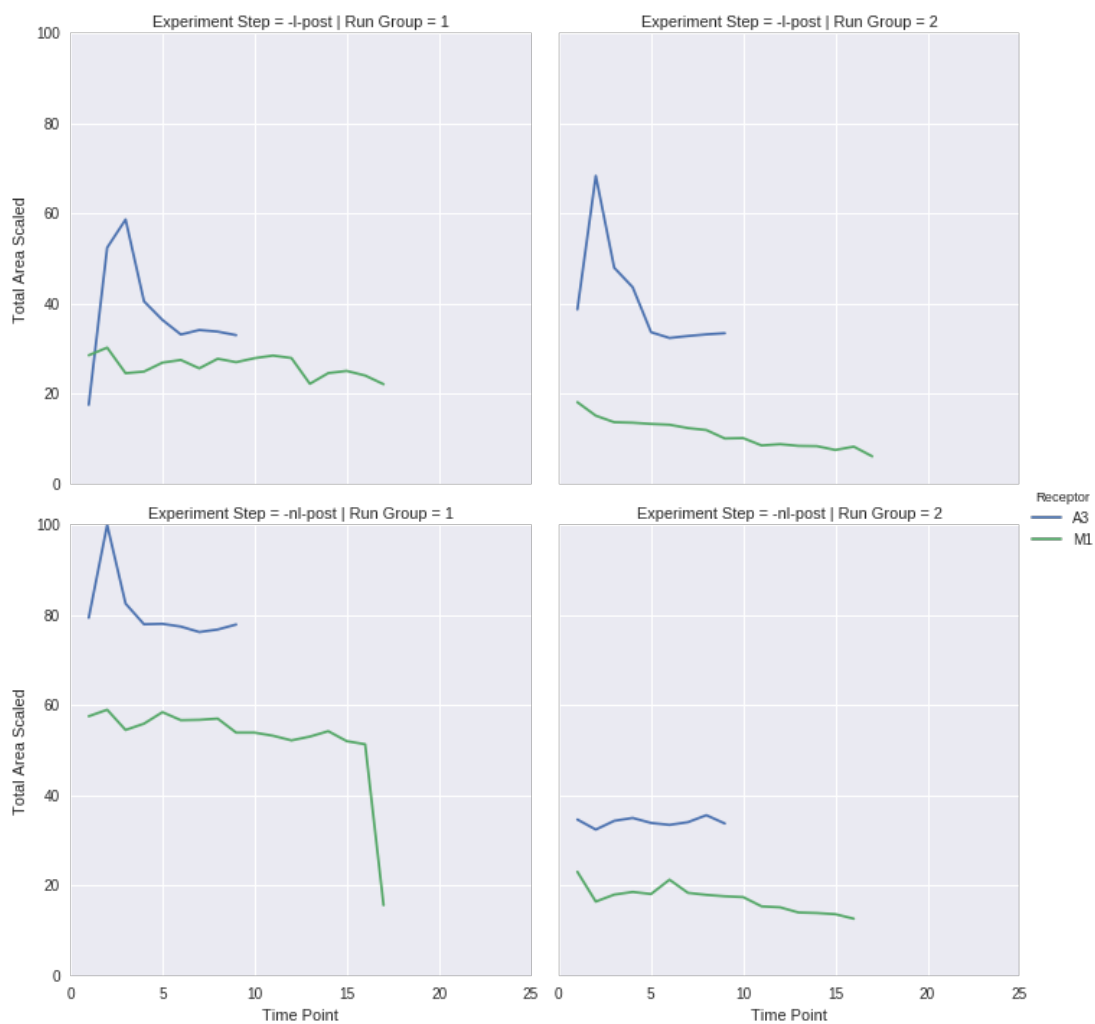
In [21]: post = ['-l-post', '-nl-post']
receptors = ['A3', 'M1']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                               (totals65['Experiment Step'].isin(post))
                               ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,25),
                  aspect=1
                  )

g.map(plt.plot, 'Time Point', 'Total Area Scaled')
g.add_legend()
g.savefig('total65_a3-and-m1_post-steps_time-series-for-area_both-run-groups
.png')

```



Receptor A3 and M1, post steps, time series for number, both run groups

```

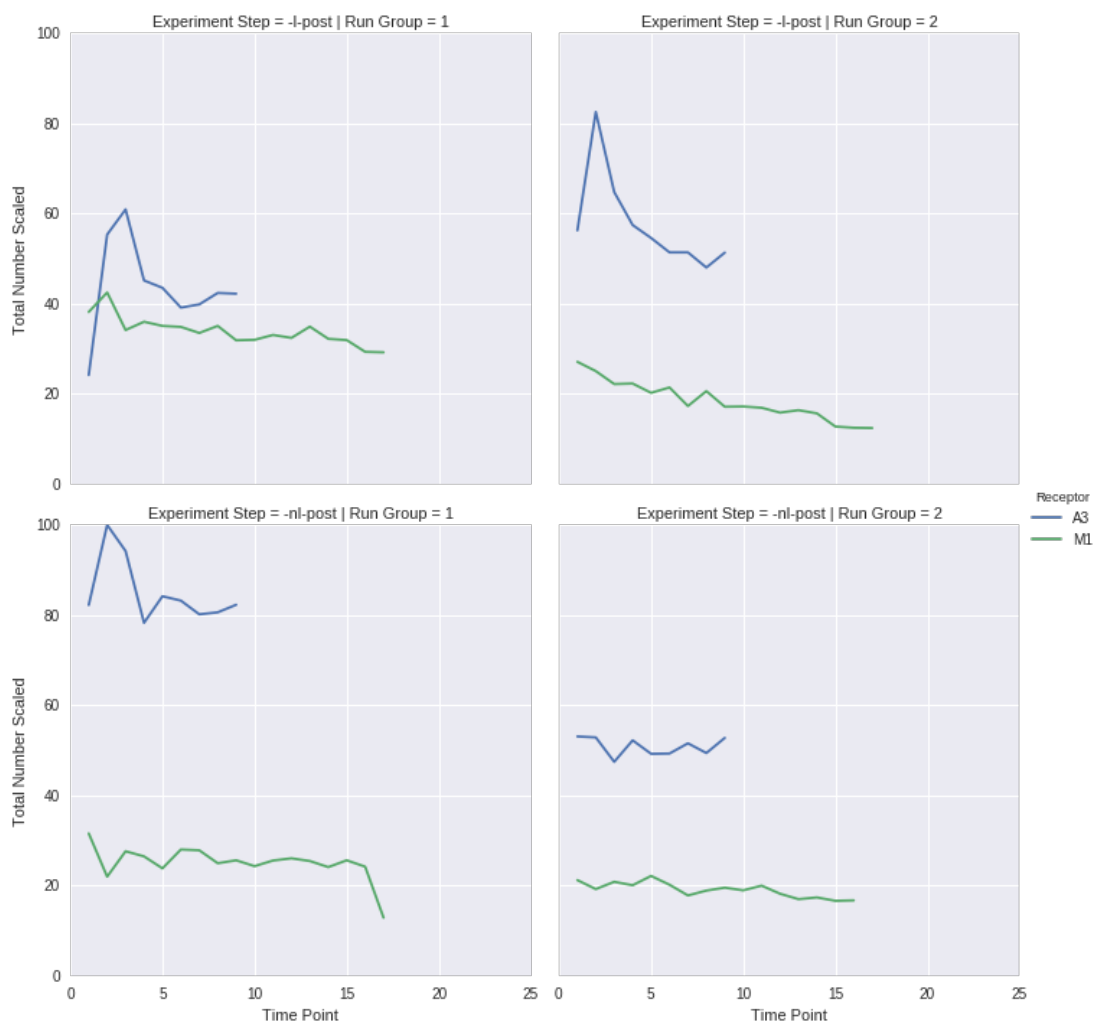
In [22]: post = ['-l-post', '-nl-post']
receptors = ['A3', 'M1']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                               (totals65['Experiment Step'].isin(post))
                               ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,25),
                  aspect=1)

g.map(plt.plot, 'Time Point', 'Total Number Scaled')
g.add_legend()
g.savefig('total65_a3-and-m1_post-steps_time-series-for-number_both-run-groups.png')

```



Receptor A3 and M1, post steps, time series for intensity, both run groups

```

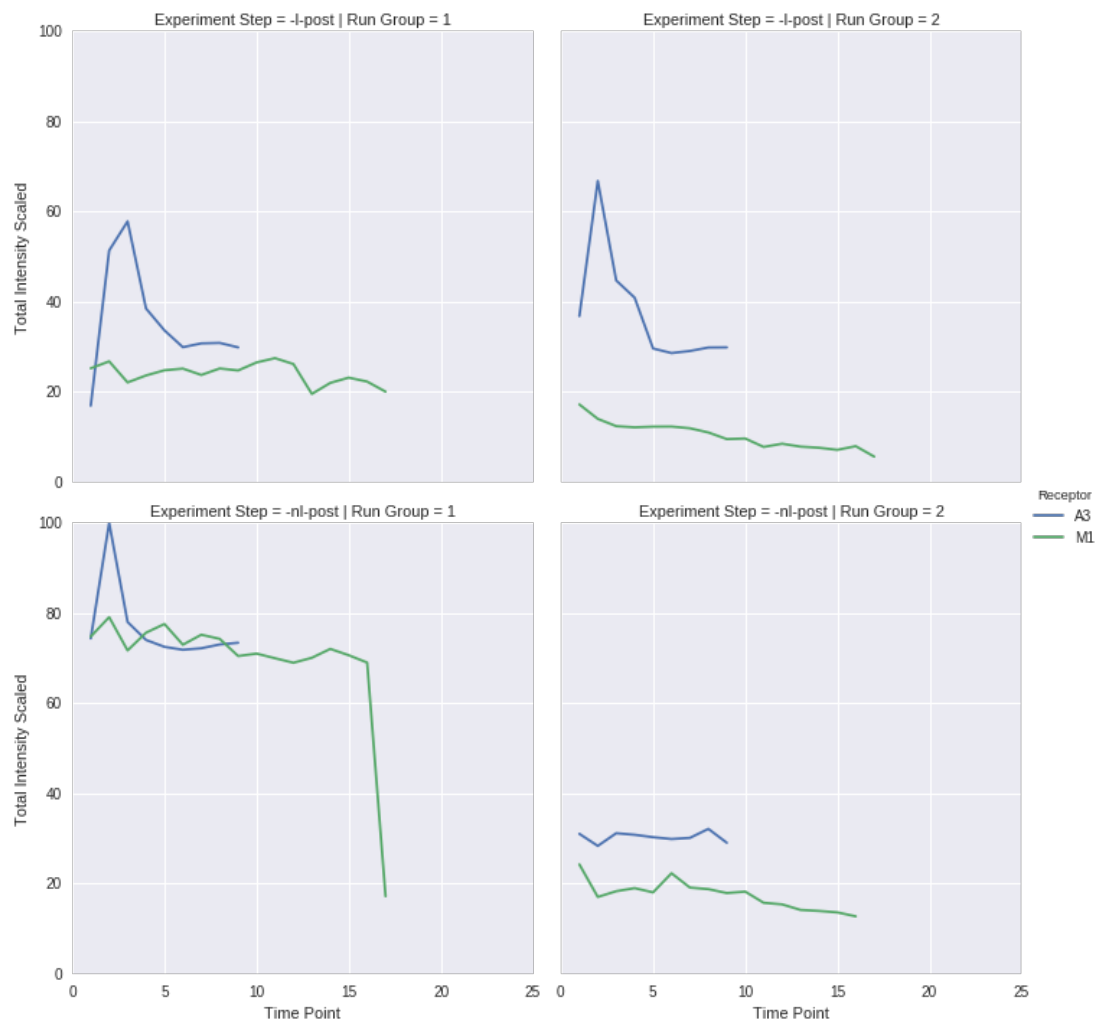
In [23]: post = ['-l-post', '-nl-post']
receptors = ['A3', 'M1']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                               (totals65['Experiment Step'].isin(post))
                               ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,25),
                  aspect=1
                  )

g.map(plt.plot, 'Time Point', 'Total Intensity Scaled')
g.add_legend()
g.savefig('total65_a3-and-m1_post-steps_time-series-for-intensity_both-run-g
roups.png')

```



Receptor A5 and mutants, post steps, time series for area, both run groups

```

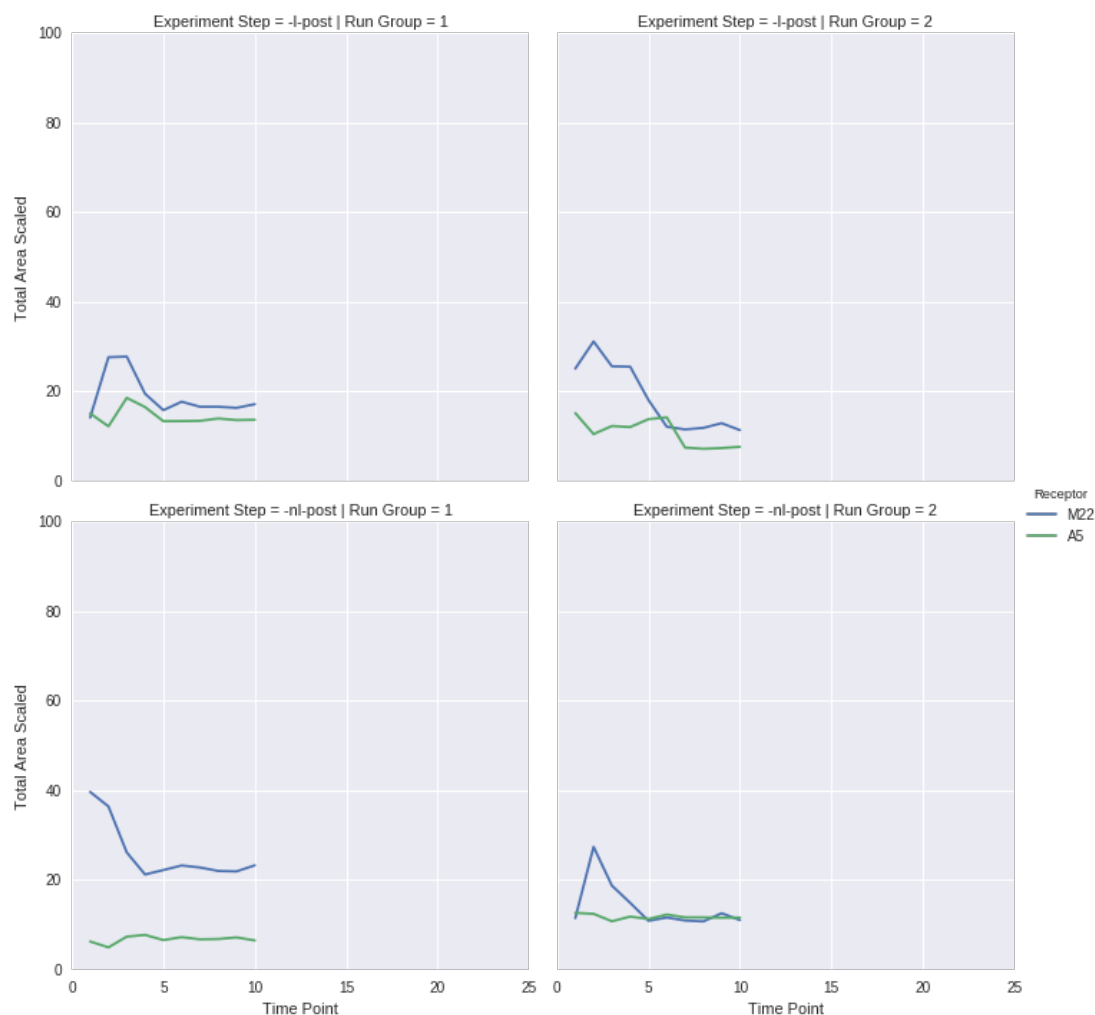
In [24]: post = ['-l-post', '-nl-post']
receptors = ['A5', 'M22']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                               (totals65['Experiment Step'].isin(post))
                               ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,25),
                  aspect=1
                  )

g.map(plt.plot, 'Time Point', 'Total Area Scaled')
g.add_legend()
g.savefig('total65_a5-and-mutants_post-steps_time-series-for-area_both-run-g
roups.png')

```



Receptor A5 and mutants, post steps, time series for number, both run groups

```

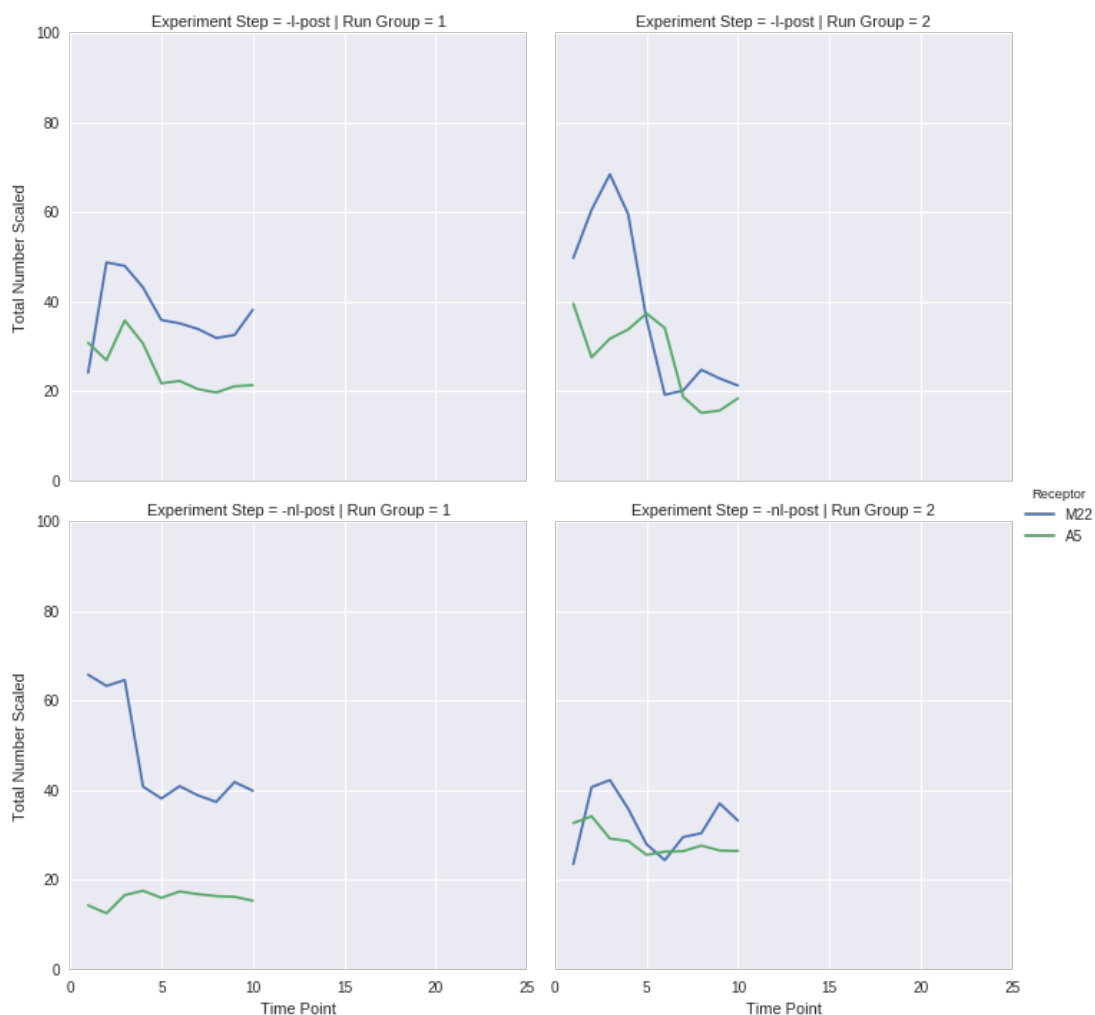
In [25]: post = ['-l-post', '-nl-post']
receptors = ['A5', 'M22']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                               (totals65['Experiment Step'].isin(post))
                               ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,25),
                  aspect=1
                  )

g.map(plt.plot, 'Time Point', 'Total Number Scaled')
g.add_legend()
g.savefig('total65_a5-and-mutants_post-steps_time-series-for-number_both-run-groups.png')

```



Receptor A5 and mutants, post steps, time series for intensity, both run groups

```

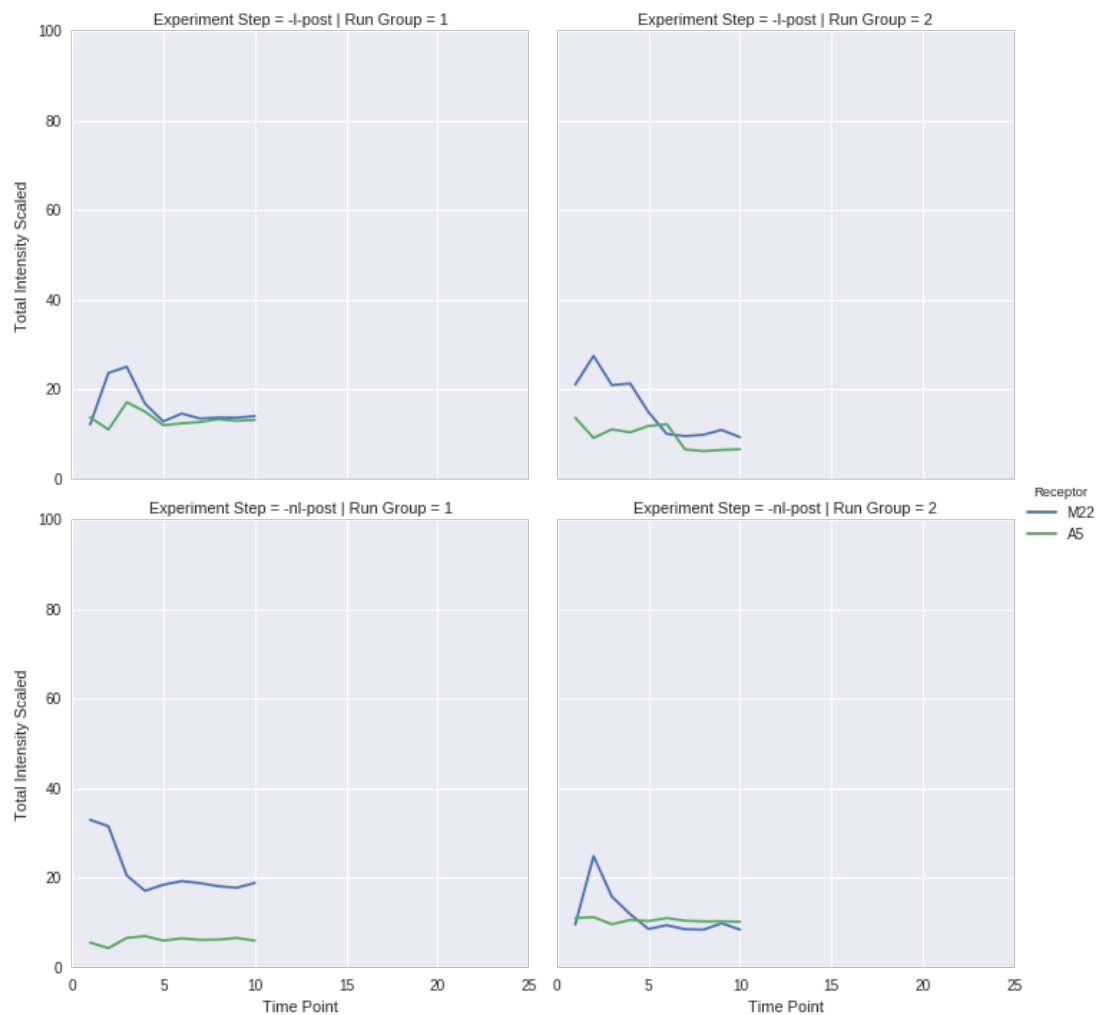
In [26]: post = ['-l-post', '-nl-post']
receptors = ['A5', 'M22']

tmpdf = None
tmpdf = pd.DataFrame(totals65[(totals65['Receptor'].isin(receptors)) &
                               (totals65['Experiment Step'].isin(post))
                               ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,25),
                  aspect=1
                  )

g.map(plt.plot, 'Time Point', 'Total Intensity Scaled')
g.add_legend()
g.savefig('total65_a5-and-mutants_post-steps_time-series-for-number_both-run-groups.png')

```



Summary statistics

```

In [27]: def print_header(title):
          print(title)
          print('Receptor    Run Group      Min      Max      Avg      Std  ')
          print('-----')

def sorted_cleanly(l):
    convert = lambda text: int(text) if text.isdigit() else text
    alphanum_key = lambda key: [ convert(c) for c in re.split('([0-9]+)', key)]
    return sorted(l, key = alphanum_key)

sorted_receptors = sorted_cleanly(set(totals65['Receptor']))
columns = ['Total Number of Clusters', 'Total Area of Protein Clusters', 'Total Intensity of Protein Clusters']

for c in columns:
    print_header(c)
    for rptr in sorted_receptors:
        for r in sorted(set(totals65['Run Group'])):
            tmpdf = totals65[ (totals65['Receptor'] == rptr) &
                              (totals65['Run Group'] == r)
                              ]
            print('{:<10}{:^9}{:>8}{:>10}{:>12}{:>12}'.format(rptr,
                                                              r,
                                                              min(tmpdf[c]),
                                                              max(tmpdf[c]),
                                                              round(np.average(tmpdf[c]),1),
                                                              round(np.std(tmpdf[c]),1),
                                                              )
                  )
    print()
    print()

```

Total Number of Clusters					
Receptor	Run Group	Min	Max	Avg	Std

A3	1	446	1845	1186.7	412.0
A3	2	875	1523	999.8	146.0
A5	1	231	718	408.9	145.5
A5	2	280	730	521.5	118.9
B1	1	275	1034	610.1	264.5
B1	2	561	1827	1204.7	441.1
M1	1	237	784	550.3	102.7
M1	2	229	500	352.9	62.8
M5	1	691	956	795.5	87.6
M5	2	638	1584	1126.1	303.1
M7	1	928	1214	1064.0	92.0
M7	2	861	1255	1023.6	94.0
M22	1	446	1298	850.4	236.0
M22	2	355	1486	745.5	326.1
M26	1	139	437	287.1	91.3
M26	2	339	783	479.2	102.1

Total Area of Protein Clusters					
Receptor	Run Group	Min	Max	Avg	Std

A3	1	4647	26460	15670.5	6202.4
A3	2	8571	18093	9868.4	2256.8
A5	1	1303	5624	2950.0	1363.8
A5	2	1898	4009	2977.8	541.7
B1	1	864	7197	2846.8	1663.3
B1	2	2557	11828	6511.0	2687.0
M1	1	4132	16953	10642.8	4005.9
M1	2	1621	6374	3851.8	1177.3
M5	1	7406	13051	9079.4	1423.4
M5	2	10208	25176	16833.9	4788.0
M7	1	7827	20513	11907.1	3657.7
M7	2	8499	21373	18976.7	3167.1
M22	1	3745	10486	6629.7	2002.7
M22	2	2836	9611	5222.9	2344.8
M26	1	1054	6496	3422.5	1863.9
M26	2	2167	6618	3866.9	902.6

Total Intensity of Protein Clusters					
Receptor	Run Group	Min	Max	Avg	Std

A3	1	508071	3003110	1683377.5	688269.4
A3	2	850631	2005959	1015754.6	272891.3
A5	1	129463	568744	306653.2	142995.4
A5	2	186310	409479	301294.2	54103.9
B1	1	69124	649347	242163.2	145785.7
B1	2	223094	1077019	591020.8	252293.2
M1	1	515196	2493810	1429516.2	749040.0
M1	2	168856	781433	433465.9	150665.6
M5	1	737176	1421679	956557.9	165288.7
M5	2	1148385	2843448	1877166.5	532849.0
M7	1	804249	2500504	1324958.0	502337.1
M7	2	854419	2834724	2414324.9	485173.4
M22	1	364604	989565	635939.0	203232.6
M22	2	253246	945624	503787.4	241533.0
M26	1	100618	728120	371261.6	215838.4
M26	2	196007	673348	381342.9	94507.3

```

In [28]: def print_header(title):
          print(title)
          print('Receptor   Run Group   Steps           Min           Max           Avg
Std   ')
          print('-----')
          print('----- ')

def sorted_cleanly(l):
    convert = lambda text: int(text) if text.isdigit() else text
    alphanum_key = lambda key: [ convert(c) for c in re.split('([0-9]+)', key)]
    return sorted(l, key = alphanum_key)

sorted_receptors = sorted_cleanly(set(totals65['Receptor']))
columns = ['Total Number of Clusters', 'Total Area of Protein Clusters', 'Total Intensity of Protein Clusters']

for c in columns:
    print_header(c)
    for rptr in sorted_receptors:
        for r in sorted(set(totals65['Run Group'])):
            tmpdf = totals65[ (totals65['Receptor'] == rptr) &
                               (totals65['Run Group'] == r)
                               ]
            print('{:<10}{:^9}{:>8}{:>10}{:>10}{:>12}{:>12}'.format(rptr,
                                                                    r,
                                                                    int(tmpdf.groupby(totals65['Receptor']).size
                                                                    ()),
                                                                    min(tmpdf[c]),
                                                                    max(tmpdf[c]),
                                                                    round(np.average(tmpdf[c]),1),
                                                                    round(np.std(tmpdf[c]),1)
                                                                    )
                                                                    )
            print()
            print()

```

Total Number of Clusters						
Receptor	Run Group	Steps	Min	Max	Avg	Std

A3	1	18	446	1845	1186.7	412.0
A3	2	18	875	1523	999.8	146.0
A5	1	26	231	718	408.9	145.5
A5	2	26	280	730	521.5	118.9
B1	1	46	275	1034	610.1	264.5
B1	2	46	561	1827	1204.7	441.1
M1	1	38	237	784	550.3	102.7
M1	2	37	229	500	352.9	62.8
M5	1	24	691	956	795.5	87.6
M5	2	24	638	1584	1126.1	303.1
M7	1	14	928	1214	1064.0	92.0
M7	2	14	861	1255	1023.6	94.0
M22	1	26	446	1298	850.4	236.0
M22	2	26	355	1486	745.5	326.1
M26	1	44	139	437	287.1	91.3
M26	2	44	339	783	479.2	102.1

Total Area of Protein Clusters						
Receptor	Run Group	Steps	Min	Max	Avg	Std

A3	1	18	4647	26460	15670.5	6202.4
A3	2	18	8571	18093	9868.4	2256.8
A5	1	26	1303	5624	2950.0	1363.8
A5	2	26	1898	4009	2977.8	541.7
B1	1	46	864	7197	2846.8	1663.3
B1	2	46	2557	11828	6511.0	2687.0
M1	1	38	4132	16953	10642.8	4005.9
M1	2	37	1621	6374	3851.8	1177.3
M5	1	24	7406	13051	9079.4	1423.4
M5	2	24	10208	25176	16833.9	4788.0
M7	1	14	7827	20513	11907.1	3657.7
M7	2	14	8499	21373	18976.7	3167.1
M22	1	26	3745	10486	6629.7	2002.7
M22	2	26	2836	9611	5222.9	2344.8
M26	1	44	1054	6496	3422.5	1863.9
M26	2	44	2167	6618	3866.9	902.6

Total Intensity of Protein Clusters						
Receptor	Run Group	Steps	Min	Max	Avg	Std

A3	1	18	508071	3003110	1683377.5	688269.4
A3	2	18	850631	2005959	1015754.6	272891.3
A5	1	26	129463	568744	306653.2	142995.4
A5	2	26	186310	409479	301294.2	54103.9
B1	1	46	69124	649347	242163.2	145785.7
B1	2	46	223094	1077019	591020.8	252293.2
M1	1	38	515196	2493810	1429516.2	749040.0
M1	2	37	168856	781433	433465.9	150665.6
M5	1	24	737176	1421679	956557.9	165288.7
M5	2	24	1148385	2843448	1877166.5	532849.0
M7	1	14	804249	2500504	1324958.0	502337.1
M7	2	14	854419	2834724	2414324.9	485173.4
M22	1	26	364604	989565	635939.0	203232.6
M22	2	26	253246	945624	503787.4	241533.0
M26	1	44	100618	728120	371261.6	215838.4
M26	2	44	196007	673348	381342.9	94507.3

